An extremal even unimodular lattice in dimension 112: supplement

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This supplement records search motivation, additional checks, running times and the support-module argument. Section, proposition, appendix and bibliography identifiers without an S prefix, and computation labels [C1]–[C31], refer to the main paper. Theorem A is proved in §§2–6 of that paper. Route T is incomplete: generation and full integer verification of the certificates for test (2) in §S4 remain unfinished. This supplement reports the completed checks without assuming that the outstanding work has been completed.

S1. Siegel means and search motivation

Write $\Sigma(n)$ for the Siegel weighted mean number of nonzero vectors of norm below the extremal bound $\mu(n)$. It is the sum of the corresponding Eisenstein-series coefficients below.

As a measure of how hard it is to find an extremal lattice, we use the average over all even unimodular lattices. By Siegel's theorem this average is an explicit modular form (an Eisenstein series), so its coefficients can be computed exactly.

Up to isometry there are finitely many even unimodular lattices of dimension n (they form a single genus). Weighting each lattice Λ by $1/|\mathrm{Aut}\ \Lambda|$, Siegel's theorem (a special case of the Siegel–Weil formula) gives

$$\frac{\sum_\Lambda \theta_\Lambda/|\mathrm{Aut}\ \Lambda|}{\sum_\Lambda 1/|\mathrm{Aut}\ \Lambda|} = E_{n/2}(q) = 1 - \frac{2k}{B_k}\sum_{m\geq 1}\sigma_{k-1}(m)\,q^m \qquad (k = n/2)$$

[Kin03, Thm. 18]. Here B_k is the Bernoulli number (negative for $k \equiv 0 \bmod 4$) and $\sigma_{k-1}(m)$ the sum of the $(k-1)$-th powers of the divisors of m. The coefficient of q^m is the average number of vectors of norm $2m$. The weight $1/|\mathrm{Aut}\ \Lambda|$ can be read as the likelihood of obtaining Λ from a "random" construction (lattices with more symmetry are less likely).

The difference between the extremal theta series and $E_{n/2}$ is a cusp form. Table S1 compares the coefficients in dimensions 56 and 112.

Table S1. Numbers of vectors by norm: Siegel mean (coefficients of $E_{n/2}$) versus the extremal theta series.

n	norm	typical lattice (Siegel mean)	extremal lattice	comment
56	2	2×10^{-6}	0	inner shell
	4	275.3	0	inner shell ($\Sigma(56) \approx 275$)
	6	15 643 264.9	15 590 400	kissing number 0.3 % below the mean
	8	36 955 007 141.0	36 957 286 800	
	10	15 284 232 608 011.0	15 284 192 071 680	
112	2	4×10^{-29}	0	inner shell
	4	1.4×10^{-12}	0	inner shell
	6	0.0069	0	inner shell
	8	51 014.2	0	inner shell; the main contribution
	10	10 907 925 307.1	10 888 335 360	kissing number 0.18 % below the mean
	12	247 007 811 617 624	247 006 775 232 000	

The mean number of nonzero vectors of norm at most 6 in dimension 112 is about 0.007; the mean number of norm-8 vectors is about 51 000. Thus most of the mean count below the extremal bound comes from norm 8. These are weighted averages over lattices. §S1.1 explains how they were used to motivate the search without interpreting them as the distribution of lattices with a prescribed symmetry group.

By the formula above, with $n = 2k$ the average number of vectors of norm $2m$ is $-\frac{2k}{B_k}\sigma_{k-1}(m)$; $\Sigma(n)$ is its sum over $1 \le m \le \lfloor n/24 \rfloor$.

Table S2. Siegel means $\Sigma(n)$ of the number of inner-shell vectors (the Siegel sawtooth). Each row has a fixed extremal bound.

	$n \equiv 0 \pmod{24}$	$n \equiv 8$	$n \equiv 16$
$\mu = 4$	24: 94.8	32: 4.51	40: 0.076
$\mu = 6$	48: 4 651	56: 275	64: 9.09
$\mu = 8$	72: 262 716	80: 16 801	88: 716
$\mu = 10$	96: 15 732 086	104: 1 046 569	**112: 51 014**
$\mu = 12$	120: 9.73×10^8	128: 6.63×10^7	136: 3.51×10^6

Within a row, $\Sigma(n)$ drops by a factor of 15–20 for every 8 dimensions. This is consistent with an extremal lattice in dimension 80 (1998) having been found before one in dimension 72 (2010); since dimension 88 followed only in 2026, this is only a rough guide.

Table S3. Rounded Siegel means of the number of nonzero vectors below the extremal bound. The last column gives a rounded group-order scale for the illustrative benchmark $\frac{\Sigma(n)}{|}|G| \le 0.05$; this is not a criterion for extremality.

n	$\mu(n)$	inner shells	$\Sigma(n)$	$\lvert G\rvert$ for $\lambda \le 0.05$
72	8	norms 2–6	262 716	5×10^6
88	8	norms 2–6	716	1×10^4
96	10	norms 2–8	15 732 086	3×10^8
104	10	norms 2–8	1 046 569	2×10^7
112	**10**	norms 2–8	**51 014**	$\mathbf{1 \times 10^6}$
120	12	norms 2–10	972 746 306	2×10^{10}

S1.1. Symmetry and the mean vector count

Let $\Sigma(n)$ be the Siegel weighted mean defined in §S1. For a chosen finite symmetry group G, consider the ratio $\lambda = \frac{\Sigma(n)}{|G|}$. If the stabilizers of the relevant vectors are trivial, their orbits have size $|G|$. This suggests using the ratio to compare the scale of the mean vector count with an orbit size. The weighted mean over all even unimodular lattices does not, by itself, give the conditional mean over lattices carrying a specified group.

As a further modeling assumption, a Poisson number of orbits with mean λ would give the probability $e^{-\lambda}$ for having no such orbit. With only the scalar symmetries ± 1, the corresponding scale in dimension 112 is $e^{-\frac{\Sigma(112)}{2}} \approx e^{-25\,507}$. This model helped motivate the imposition of symmetry during the search; it is not a deduction about the actual distribution of lattices, and it estimates neither the correctness nor the running time of the proof.

Table S4. The ratio $\lambda = \frac{\Sigma(n)}{|G|}$ for selected lattices. $|G|$ is the order of the automorphism group (for gluings A, B: of the visible subgroup; for L_{112}: Theorem B, §8).

lattice	$\Sigma(n)$	$\lvert G\rvert$	λ	how it was found
P_{48n}	4 651	524 160	0.009	from a group
P_{48p}	4 651	145 728	0.03	from a code
P_{48q}	4 651	103 776	0.04	from a code
P_{48m}	4 651	1 200	3.9	exhaustive classification
Γ_{72}	262 716	5 241 600	0.05	test of all 9 structures
L_{80} (Bachoc–Nebe)	16 801	4.5×10^9	4×10^{-6}	from a group
gluings A, B (min 8)	51 014	292 320	0.17	§3
gluing C = L_{112}	51 014	1 169 280	**0.044**	this paper
(L_{112}, counting $\mathrm{SL}_2(29)$ only)	51 014	24 360	2.1	

Group orders are taken from [NS], with Γ_{72} from [Neb12]. The last two rows illustrate that the numerical ratio changes substantially depending on whether one counts only the imposed group or the full automorphism group determined later. The proof uses the imposed group and does not require a small value of this ratio.

S2. Expected collisions in a random gluing

If f were a random parity-preserving map, the class xf of $x \in L_{56}(a)$ would be uniformly distributed among classes of the same parity. Since each class contains at most one pair $\pm v$ of norm b (a consequence of §5), the expected number of inner-shell vectors of L_{112} coming from type (a, b) is

$$\mathbb{E}[\#(a,b)] \approx \frac{|L_{56}(a)|\,|L_{56}(b)|}{N_\pm}, \qquad N_- = 2^{55} - 2^{27} \text{ (odd classes)}, \quad N_+ = 2^{55} + 2^{27} \text{ (even classes)}$$

(computation in §S2.1). This estimate is a heuristic and is not used in the proof.

Table S5. Expected numbers for a random gluing (from the theta series of L_{56}) and the actual values for gluings A, B, C.

type	expected	share	gluings A, B	gluing C
$(6,6)$	0.007	—	0	0
$(6,10)$ and $(10,6)$	13 228	26 %	0	0
$(8,8)$	37 910	74 %	292 320	0
total	51 137		292 320	0

The total agrees with the Siegel mean $\Sigma(112) = 51\,014$ within 0.2 %. Counting both inputs and partners modulo sign gives the probability scale $\frac{6\,\frac{614}{4}}{15\,590\,\frac{400}{2}} \approx \frac{1}{4}\,714$. Exhaustive tests are therefore required.

S2.1. Calculation of the expectations

Type $(6,10)$: $|L_{56}(6)||L_{56}(10)|/N_- = 15\,590\,400 \times 15\,284\,192\,071\,680/(2^{55}-2^{27}) \approx 6\,614$; type $(10,6)$ gives the same, together 13 228. Type $(8,8)$: $|L_{56}(8)|^2/N_+ \approx 37\,910$. Type $(6,6)$: $|L_{56}(6)|^2/N_- \approx 0.0067$. All three values assume that the class xf is uniformly distributed among classes of the same parity.

S3. Reproduction and additional checks

S3.1. Running times

Table S6. Main computations (single core unless stated).

step	method	cost
F1	interval enumeration, norm ≤ 4, BKZ-40 basis	34 min (8.5×10^9 nodes); floating-point enumerations 51 min – 2.3 h
F2	closure + orbits	about 13 min (finding 10 min, checking 3 min)
F3	marked sums of two minimal vectors	about 1.2 h (checking 9 min)
F5a	odd-coset certificates (656)	finding 4 min, checking 2 min
F5b	canonical forms (two implementations)	24 min – about 2 h
F5b	CVP, radius² 2.05	4 × 9 h
F5b	even-coset certificates	finding 6 × about 31 h (shared machine); checking 7.9 h on 3 threads

The computations were run with Python 3.12.15, NumPy 2.5.4, SageMath 10.9, fplll 5.5.0 (fpylll 0.6.4), PARI/GP 2.17.3 and GAP 4.15.1 [Soft]. Recorded checks in a second environment used Debian 13.7, Python 3.13.5, NumPy 2.5.4 and gcc 14.2. The times above reflect the machine load at the time of measurement; each verification program records its Python, NumPy and OS versions and running time in JSON. The Lean version is 4.34.1, with mathlib pinned to revision `d13f23b723b8a846827a245b89c10fc7d3f11612`.

S3.2. Verification routes

An independent verification needs only F0 (L_{112} is even unimodular) and F1–F5; the search (failed constructions, classification of gluings, choice of τ) is not needed. Each of F0–F5 can be checked in several ways, and the combination determines which criteria and programs are used and how long it takes. We do not designate one route as official; Table S7 lists them. The inputs and criteria for each route are listed below.

Baseline. Enumerating all vectors of norm ≤ 8 of L_{112} itself (dimension 112) would need about 10^{25} enumeration nodes by the Gaussian volume estimate (5.3×10^{25} with an LLL basis, 4.1×10^{24} with a BKZ-30 basis) [C16] at 10^7 nodes per second this would take 10^{18} seconds, which is infeasible.

Table S7. Verification routes. [m] measured; [e] estimated from measured per-operation costs.

route	content	criteria and programs used	total
A (fastest)	F1 by interval enumeration (34 min [m]); F2, F3 by orbit sizes and distinct canonical forms (3 min and 9 min [m]); F5a by odd-coset certificates (2 min [m]); F5b by canonical-form look-up (24 min [m])	the interval enumerator, the canonical-form code, Proposition 6.3	about 1.2 CPU hours [m]
B (no canonical forms in the decision)	as A, with F5b by checking the distributed even-coset certificates (7.9 h on 3 threads [m]); finding them again would take 763 992 pairs (using the frame-preserving symmetry [C14]) $\times$ 0.7 s [m]	the interval enumerator, the midpoint lemma	about 25 CPU hours to check [m]; about 150 to find again [e]
C (closest vectors)	as A, with F5b by closest-vector enumeration	pruning in the enumeration (not a proof if floating point)	about 40 CPU hours [m]
D (from scratch)	rebuild the orbit lists from the four data files only (the approach of implementation 3)	one's own code only	A–C plus 2–3 hours [e]
S ([C15], sampling; **not a proof**)	re-check k random certificates with code independent of the producer (group rebuilt from the generators, the distributed τ, 64-bit integers only)	—	$k = 200 + 200$ in 8 min [m]; **all 400 pass**
T (support modules)	classify the rank-2 support modules that could carry a vector of norm ≤ 8 (97 invariant types, 63 dangerous) and exclude them by three closest-vector searches modulo 2 (§S4.1)	Coulangeon's inequality; the classification (two independent enumerations agree); the searches (floating point; integer certificates and full verification remain incomplete)	classification 1 min; searches 1.6 h + 3 $\times$ 13 h + F5b [m]

Route S is a limited consistency check on the supplied tables: if a fraction p of entries were wrong, a sample of $k = 200$ would detect it with probability $1-(1-p)^{200}$, which is 0.87 for $p = 10^{-2}$ but only 0.001 for $p = 6.5 \times 10^{-6}$ (the rate of collisions in gluing B). Route T uses the Hermitian structure of $L_{112} = H \otimes P$ for the rank-2 case and the argument of §S4.2 for dependent components. It shares the norm-8 exclusion with the main proof. The extent to which the closest-vector searches have been replaced by checked representatives is stated in §S4.1. The Lean formalization in §7.1 covers the common deduction from the computational statements to the main conclusion, not the support-module enumeration.

Search and verification. Table S6 gives the measured costs of producing and checking each finite data set. A lower-bound verification must check the complete orbit lists and all relevant classes; a counterexample, in contrast, is verified by checking the membership and norm of one supplied vector.

S3.3. Recomputation by separately written implementations

Implementation 1 constructed L_{112} and performed the main computations. Implementation 2, sharing only data files (Gram matrices, generators, τ, orbit representatives), re-implemented every step: even unimodularity and the group action on the 112-dimensional Gram matrix, the minimum of L_{56} with different software and basis, disjointness and design identities for the orbit lists, the canonical-form test, the check of the certificates, and a sample of 180 classes by exhaustive enumeration in the lattice $\langle 2L_{56}, t\rangle$ with PARI. Implementation 3 used only the four data files of Appendix D and wrote all of its code from scratch (PARI/GP, fplll, GAP, C). It reproduced the orbit counts with the same completeness proof, found no collision of any type for L_{112}, found exactly 6 colliding orbits (292 320 vectors of norm 8) for the other class of τ, and reproduced the 112-dimensional Gram matrix as PARI's `qflllgram` of (3.1). Its enumerations also prune in floating point.

S3.4. Changes of frame

Write $L_{112} = L_{56} \otimes_{M'} \Lambda$ with $\Lambda = \{(a,b) \in M'^2 : b \equiv a\tau \bmod 2M'\}/\sqrt{2}$, a Hermitian M'-lattice of real dimension 16. A **frame** is a pair (w_1, w_2) of orthogonal elements of Λ of Hermitian norm 2. Each frame exhibits $L_{112} \supset L_{56}w_1 \perp L_{56}w_2 \cong \sqrt{2}L_{56} \perp \sqrt{2}L_{56}$ and thus L_{112} as a gluing L_φ; the class of φ in Proposition 3.3(c) determines the minimum, so it must be the same for all frames. Λ has 72 elements of norm 2 and 864 frames. For τ of class C all 864 frames give class C, and for class B all give class B [C17]. This test is inexpensive and independent of the collision computations.

S3.5. Fixed sublattices and BKZ

Elements of order 3, 5, 7 of $\mathrm{SL}_2(29)$ have fixed sublattices of L_{112} of dimension 32, 16, 16 with exact minima 12, 10, 14 (determinants $3^{32}, 5^{16}, 7^{16}$) [C18]. The fixed sublattice of an element of order 5 contains explicit vectors of norm 10, so $\min L_{112} \le 10$. These types are among those not excluded for extremal lattices of dimension 112 by [Neb13]. These fixed sublattices do not establish a lower bound for the minimum of the whole lattice: gluing B has the same fixed-sublattice minima while its own minimum is 8. Likewise BKZ reduction of L_{112} with block size up to 60 finds no vector below norm 12, but also fails to detect a norm-8 vector in B.

S3.6. Controls and closest-vector searches

Every exhaustive test above was also run on the positive controls with known answers (Table 6). In particular, the tests detect *rare* collisions: 10 out of 1 527 984 pairs for B. The CVP test was cross-checked against the canonical-form test on all colliding pairs and on a sample of 150 pairs.

CVP. The class $t + 2L_{56}$ ($t = x_i u\tau$) contains a vector of norm ≤ 8 iff some $y \in L_{56}$ satisfies $\langle y + t/2, y + t/2\rangle \leq 2$, a closest-vector problem in L_{56} with target $-t/2$. We ran fplll's enumeration at squared radius 2.05 for all 1 527 984 pairs (four jobs of about 9 hours, without canonical forms or K_8): **0 collisions** [C12]. The same enumeration found the norm-8 partner in every one of the 46 colliding pairs of gluings A and B. The radius can be reduced to this value because a norm-8 element corresponds to squared distance exactly 2, and the next possible norm, 12, to 3.

S3.7. Scope of the checks

Table S8. Possible errors and the checks that would detect them.

possible error	would be detected by
Orbit lists incomplete (shared by all $(8,8)$ tests)	orbit-size sums (exact); recount from K_8; design identities of degree 2, 4, 6; implementation 3 from scratch
The canonical-form criterion is wrong	Proposition 6.3 (proof); CVP test without canonical forms; even-coset certificates for all pairs; support-module route (§S4.1)
A gap in the logic (missing type, case $x = 0$, wrong direction)	written proofs (§§3–5); three implementations re-deriving the reduction; tests in both directions τ, τ^{-1}; **formalization in Lean (§7.1)**
The 112-dimensional Gram matrix is not the lattice described	two implementations rebuilt (3.1) independently; the third reproduced the LLL-reduced matrix exactly; frame test
$\min L_{56} < 6$	**rigorous interval enumeration** reduction to $L_{56,2}(M)$; three floating-point enumerations; exact agreement of shell sizes and design identities (indirect)
Assignment of minima to classes inconsistent	frame-change test (864 frames)
The tests are blind to rare collisions	positive controls A, B (10 in 1.5 million detected)
An error in a shared computational hypothesis	the corresponding input, completeness and arithmetic checks in §§6–7; Lean checks the deduction from these hypotheses, not their computational verification (§7.1)

S4. Support modules (route T)

S4.1. Rank-two support modules

Gluings B and C use the same base lattice and differ only in the choice of τ. For a vector $z = e_1 \otimes x + e_2 \otimes y$ of $H \otimes P$ define its **support module** $N_z = \mathcal{O}x + \mathcal{O}y \subset P$. In this section we use the trace-normalized Hermitian form $k_P = h_P/2$, for which $\langle x, x\rangle = \mathrm{Tr}_{K/\mathbb{Q}} k_P(x,x) = 2k_P(x,x)$, and $k_R = h_H \otimes k_P$; determinants $\det_k$ of support modules and Table S9 refer to k_P. Coulangeon's inequality for tensors of rank 2 gives $k_R(z,z) \geq 2\sqrt{\det h_H \cdot \det_k N_z}$ [Cou00, CN13], so a vector of norm 8 ($k_R(z,z) = 4$) needs $\det_k N_z \leq 4$. For 20 symmetry-inequivalent norm-8 vectors of B the support modules are of two kinds, both saturated and both free of minimal vectors (Table S9) [C19].

Table S9. Support modules of the norm-8 vectors of gluing B.

type	reduced Gram matrix of k_P	$\det_k$	count (of 20)
I	$\begin{pmatrix} 4 & -2\sqrt{-5} \\ 2\sqrt{-5} & 6 \end{pmatrix} = 2H^{-1}$	4 (equality)	12
II	$\begin{pmatrix} 4 & \frac{9}{5}\sqrt{-5} \\ -\frac{9}{5}\sqrt{-5} & 5 \end{pmatrix}$	19/5	8

These matrices describe the form in a K-basis of the support, rather than an integral $\mathcal{O}$-basis. In particular the rational entries for type II do not specify the integral module structure, which is retained separately.

Type I is exactly twice the inverse of H, the configuration in which Coulangeon's bound is attained. This parallels the "Barnes-free" criterion for Γ_{72} in [CN13].

Complete classification (route T). Let $z \in L_{112}$ have norm ≤ 8. If x, y are independent over K, put $N_0 = \mathcal{O}x + \mathcal{O}y$ and let $N = P \cap (Kx + Ky)$ be its saturation. Since H is free, $z \in H \otimes N_0 \subset H \otimes N$. Coulangeon's inequality $k_{R(z,z)} \geq 2\sqrt{\det h_H \cdot \det_k N_0}$ [Cou00, Lemma 3.3 and Prop. 3.2] gives $\det_k N_0 \leq 4$, and $\det_k N = \det_k N_0/[N : N_0] \leq 4$ (for Hermitian lattices the index enters to the first power; for non-free N the determinant is taken with respect to a pseudo-basis, including the norm of the coefficient ideal). So N is a saturated rank-2 submodule of P with real minimum ≥ 6 and $\det_k N \leq 4$. If x, y are dependent over K, a vector of norm ≤ 8 is a collision of type $(6,6)$ or $(8,8)$ (§S4.2), detected by the searches (3) and (1) below. Because $\mathcal{O} = \mathbb{Z}\left[\sqrt{-5}\right]$ has class number 2, N need not be free. All such N were enumerated, free and non-free, as Hermitian $\mathcal{O}$-lattices with even integral trace form, and grouped by invariants (determinant, theta coefficients up to norm 16, minimum and numbers of short vectors of $H \otimes N$): **97 invariant types, 63 of them dangerous** ($H \otimes N$ has a vector of norm < 10; 30 with minimum 6 and 33 with minimum 8). The grouping is by invariants, not by isometry, and the exclusion below was checked on every enumerated candidate, not only on one per type, so isometry classes are not needed. A second, independent enumeration (from a pseudo-basis through the true shortest vector, with exact rational Fincke–Pohst; the first starts from a basis-extendable shortest vector) uses a coarser invariant; regrouping both enumerations by that same invariant gives 58 dangerous types in each, with agreement at this coarser level [C20]. Every dangerous candidate contains a configuration that one of three closest-vector searches modulo 2 must detect: (1) a norm-8 vector x and a norm-8 element of the class $x\tau + 2L_{56}$ (this is the $(8,8)$ test of §6); (2) a norm-8 x and an element of norm ≤ 8 (zero included) of $x(1+\tau) + 2L_{56}$; (3) a norm-6 u and an element of norm ≤ 10 of $u\mu + 2L_{56}$, $\mu \in \{1, \tau, 1+\tau\}$. For L_{112}, test (1) is F5b. Test (2) enumerated **763 992** pairs in floating point without finding an element of norm at most 8 [C21]. Generation and full integer verification of representatives for these pairs remain to be completed. For (3), the inputs are the 656 norm-6 S-orbit representatives times three centralizer representatives after reduction by the symmetry preserving τ: $656 \times 3 = 1\,968$ input vectors u. Each is tested with $\mu \in \{1, \tau, 1+\tau\}$. For $\mu = 1$, $\pm u$ already belongs to the coset, and only elements independent of u over K are relevant. For the other two maps, all $1\,968 \times 2 = 3\,936$ supplied representatives have been checked in integer arithmetic. They exclude norms at most 10 for $\mu = \tau$, and at most 8 for $\mu = 1 + \tau$. The latter cosets are even, so norm 10 is excluded by parity as well [C21]. The enumeration found no dangerous independent configuration.

Completing this as a rigorous alternative route requires generation and full integer verification of every representative in (2), and a completeness check of the support-module parameter ranges and the reduction to the three searches. Theorem A does not use this route and is proved in §4.3 of the main paper. The collisions of gluing B were detected as controls.

S4.2. Dependent components

Let $z = \frac{1}{\sqrt{2}}(x, v) \in L_{112}$ with $v = x\tau + 2y$, and let $K = \mathbb{Q}(\sqrt{-5})$ act on $L_{56} \otimes \mathbb{Q}$ by $\sqrt{-5} \mapsto \tau$. Suppose $y \in Kx$ and $\langle z, z\rangle \leq 8$. By the proof of Lemma 4.2, $x \neq 0$ and $v \neq 0$, and $(n, m) = (\langle x, x\rangle, \langle v, v\rangle) \in \{(6,6), (6,10), (10,6), (8,8)\}$. We show $m = n \in \{6, 8\}$, i.e. z gives a collision of type $(6,6)$ or $(8,8)$ with

$v = \lambda x$, $N(\lambda) = 1$. (The claim that such a z has norm ≥ 12 is false in general: for a gluing along an element of the centralizer, $v = xu$ gives the forced collisions of §3.2.)

Since $\langle x, x\tau\rangle = 0$ and $\langle x\tau, x\tau\rangle = 5n$, the map $\lambda \mapsto \lambda x$ identifies K with Kx and carries the norm form to $nN(\lambda)$. The lattice $\Lambda = L_{56} \cap Kx$ is a rank-1 $\mathcal{O}$-module, $\Lambda = \mathfrak{a}x$ for a fractional ideal $\mathfrak{a} \supset \mathcal{O}$; put $k = [\mathfrak{a} : \mathcal{O}]$. As a $\mathbb{Z}$-lattice of rank 2, Λ is even with minimum ≥ 6 (F1) and determinant $5n^2/k^2$; Hermite's bound in dimension 2 gives $\det \Lambda \geq 3 \cdot 36/4 = 27$, so $k \leq 2, 3, 4$ for $n = 6, 8, 10$. The integral ideals of $\mathcal{O}$ of norm ≤ 4 are $\mathcal{O}$, $\mathfrak{p}_2 = \left(2, 1 + \sqrt{-5}\right)$, the two primes $\mathfrak{p}_3, \mathfrak{p}_3' = \left(3, 1 \pm \sqrt{-5}\right)$ and $\mathfrak{p}_2^2 = (2)$, so $\mathfrak{a} \in \left\{\mathcal{O}, \mathfrak{p}_2^{-1}, \mathfrak{p}_3^{-1}, \mathfrak{p}_3'^{-1}, \frac{1}{2}\mathcal{O}\right\}$. Write $v = \lambda x$; then $\lambda \in \sqrt{-5} + 2\mathfrak{a}$ (because $y \in \Lambda$) and $N(\lambda) = m/n \in \{1, 5/3, 3/5\}$.

- $\mathfrak{a} = \mathcal{O}$: $\lambda = 2p + (1 + 2q)\sqrt{-5}$ with $p, q \in \mathbb{Z}$, so $N(\lambda) \geq 5$. Impossible.
- $\mathfrak{a} = \frac{1}{2}\mathcal{O}$ (only $n = 10$): $1/2 \in \mathfrak{a}$ would give a vector of norm $n/4$, not an even integer. Impossible.
- $\mathfrak{a} = \mathfrak{p}_2^{-1}$: $2\mathfrak{a} = \mathfrak{p}_2$ and $\sqrt{-5} + \mathfrak{p}_2 = \mathcal{O} \setminus \mathfrak{p}_2$, whose norms are odd integers. Hence $N(\lambda) = 1$ and $m = n$.
- $\mathfrak{a} = \mathfrak{p}_3^{-1}$ or $\mathfrak{p}_3'^{-1}$ ($n \in \{8, 10\}$): $N(\lambda) \in N(\mathfrak{a})\mathbb{Z} = \frac{1}{3}\mathbb{Z}$ excludes $3/5$, so $N(\lambda) = 1$ and $(n, m) = (8, 8)$.

Such collisions are excluded by F5; on route T they are detected by the searches (3) and (1) of §S4.1.

S5. Constructions considered in the search

Before dimension 112. More than 190 candidate lattices in dimensions 96–112 (cyclotomic ideal lattices and their ideal-class twists, Hermitian tensor products over imaginary quadratic fields, gluings $48 + 48$ and 48×2) were tested first; every one had inner-shell vectors. In dimension 96 the ratio λ of the available symmetry groups was ≥ 300; in hindsight this is consistent with these failures, and it motivated the search for constructions with larger symmetry.

The first four gluings in 112 came from Hermitian tensor products $T \otimes L_{56,2}(M)$ over imaginary quadratic fields ($\mathbb{Q}(\sqrt{-11})$: one class; $\mathbb{Q}(\sqrt{-2})$: three classes); all have forced collisions of type $(6, 6)$. Generalizing these collisions gives the classification of Proposition 3.3.

Gluings A and B. Both have exactly 6 colliding S-orbits of type $(8, 8)$, giving 292 320 vectors of norm 8, which equals the order of the explicitly constructed automorphism subgroup. They are in fact isometric: enumerating the $\mathrm{SL}_2(29)$-equivariant maps between them as in §8 yields 12 equivariant isometries up to sign, one of which is distributed with the verification data and checked in integer arithmetic [C28].

Base lattices. $L_{56,2}(M)$ is taken from the catalog entry `L_56b`. The second base lattice was obtained inside it: for each singular $X \in M/2M$ of rank 28, $L_X = L_{56,2}(M)X + 2L_{56,2}(M)$ with the form halved is $\mathrm{SL}_2(29)$-invariant; five are even unimodular, one with centralizer of order 24 and four with order 12. Our L_{56} is one of the latter. The catalog entry `L_56c` ($= L_{56,2}(\widetilde{M})$) lists generators of a subgroup of order $2^3 \cdot 3 \cdot 7 \cdot 29$ only and states that the automorphism group contains $\mathrm{SL}_2(29) \circ \tilde{S}_3$, which is consistent with a centralizer of order 12.

S6. The $\mathrm{SL}_2(29)$-invariant even unimodular lattices of dimension 112

Table 2 of the main paper lists the equivariant gluings modulo 2 of two copies of one base lattice. This section determines all even unimodular lattices of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112}, without assuming any gluing construction. The result is that there are exactly eight such lattices up

to equivariant isometry; L_{112} is one of them, and all but one of the others contain a vector of norm at most 8.

Setting. Throughout, $S = \mathrm{SL}_2(29)$, $K = \mathbb{Q}\big(\sqrt{29}\big)$, and $\mathcal{Q}$ is the quaternion algebra of §2.4 with canonical involution $q \mapsto \bar{q}$, reduced norm nrd and reduced trace trd. Let $L = L_{56,2}(M)$ be Nebe's lattice for the maximal order M whose centralizer has order 24, $W = L \otimes \mathbb{Q}$, and $\tilde{L} = L_{56,2}\big(\widetilde{M}\big) = L_{56}$. The ring $M = \mathrm{End}_S(L)$ acts on W from the right; W is a right $\mathcal{Q}$-vector space of dimension 7, and the adjoint of right multiplication by q for the form of L is right multiplication by $\bar{q}$ (the adjoint involution is positive, hence the canonical one). Since L is unimodular, $Lq \subset L$ holds exactly for $q \in M$.

On $\mathcal{Q}^2$ we use the Hermitian form $h(\lambda,\mu) = \lambda_1\bar{\mu}_1 + \lambda_2\bar{\mu}_2$, and $\mathcal{Q}^2$ acts on W by $x\lambda = (x\lambda_1, x\lambda_2) \in W^2$, where W^2 carries the orthogonal sum of two copies of the form of W. Then $\langle x\lambda, y\mu\rangle = \langle x\, h(\lambda,\mu), y\rangle$ for $x, y \in W$. A *lattice* $\Lambda \subset \mathcal{Q}^2$ is a finitely generated left M-submodule spanning $\mathcal{Q}^2$; its dual is $\Lambda^h = \{\mu : h(\Lambda,\mu) \subset M\}$, Λ is *unimodular* if $\Lambda^h = \Lambda$, and $U(\Lambda)$ is the group of M-linear bijections of Λ preserving h. For a lattice Λ put $L \otimes_M \Lambda = \sum_{\lambda\in\Lambda} L\lambda \subset W^2$.

Theorem S6.1. Let X be an even unimodular lattice of rank 112 on which S acts by isometries with $X \otimes \mathbb{Q} \cong W \oplus W$ as $\mathbb{Q}S$-modules (as S acts on L_{112}). Then:

(a) $X = L \otimes_M \Lambda$ for the unimodular lattice $\Lambda = \mathrm{Hom}_S(L, X) \subset \mathcal{Q}^2$. The classes of such X up to S-equivariant isometry correspond bijectively to the isometry classes of unimodular lattices Λ for which $L \otimes_M \Lambda$ is even, and the centralizer of S in $\mathrm{Aut}(X)$ is isomorphic to $U(\Lambda)$.

(b) There are exactly 8 isometry classes of unimodular lattices $\Lambda \subset \mathcal{Q}^2$, with $|U(\Lambda)| = 1152, 12, 288, 24, 24, 20, 288, 48$ [C29]. For all of them $L \otimes_M \Lambda$ is even [C31]. Hence there are exactly 8 lattices X up to S-equivariant isometry, $X_0, \ldots, X_7$ (Table S10), and at most 8 up to isometry. Their Gram matrices, their bases in the coordinates of (3.1), and S-equivariant isometries $X_7 \to L_{112}$ and $X_3 \to$ gluing A are distributed with the verification data and checked in integer arithmetic [C30, C31].

(c) $X_0 = L \perp L$, $X_2 = \tilde{L} \perp \tilde{L}$ and $X_6 = L \perp \tilde{L}$ have minimum 6; X_3 is gluing A = gluing B, of minimum 8; X_4 and X_5 have minimum 8 (vectors of norm 8 are distributed and checked [C31]); $X_7 \cong L_{112}$ has minimum 10 (Theorem A). The minimum of X_1 is 8 or 10 and has not been decided at the time of writing.

(d) For $j \neq 7$, X_j is not isometric to L_{112}, even by an isometry that does not respect S.

Table S10. The 8 classes. n_1 is the number of $\pm$-pairs of $\lambda \in \Lambda_j$ with $h(\lambda,\lambda) = 1$. A frame b (resp. c) is a sublattice $J \perp J'$ of index 2^8 with $h(J,J), h(J',J') \subset 2M$ and $J/\sqrt{2}$, $J'/\sqrt{2}$ isometric to the rank-1 unimodular lattice M (resp. the other class I) [C30] through it X_j is a gluing modulo 2 of L (b) and $\tilde{L} = L \otimes_M I$ (c).

j	$\lvert U(\Lambda_j)\rvert$	n_1	frames	X_j	$\min X_j$
0	1152	24	bb	$L \perp L$ (Table 2, row 1)	6
1	12	0	bc	gluing of L and $\tilde{L}$ modulo 2	8 or 10 (open)
2	288	0	cc	$\tilde{L} \perp \tilde{L}$ (Table 2, row 3)	6
3	24	0	bb, cc	gluing A = gluing B	8
4	24	0	bc	gluing of L and $\tilde{L}$ modulo 2	8
5	20	0	none	not a gluing modulo 2	8
6	288	12	none	$L \perp \tilde{L}$	6
7	48	0	cc	L_{112} (gluing C)	10

Proof of (a). Let A be the image of $\mathbb{Z}S$ in $\mathrm{End}(L)$ and $E = \mathrm{End}_M(L)$, the ring of endomorphisms of L commuting with the right action of M; $A \subset E$, and E is a $\mathbb{Z}$-lattice of rank $392 = 7^2 \cdot 8$ which is pure in $\mathrm{End}_{\mathbb{Z}}(L)$. By the double centralizer theorem, $A \otimes \mathbb{Q} = \mathrm{End}_{\mathcal{Q}}(W)$, so A and E are orders in the same algebra. We claim $A_p = E_p$ for every prime p (subscripts denote $\otimes \mathbb{Z}_p$). For $p \nmid |S|$, $\mathbb{Z}_pS$ is a maximal order of $\mathbb{Q}_pS$ and A_p is its image in a simple factor, hence a maximal order of $E_p \otimes \mathbb{Q}$ contained in the order E_p; so $A_p = E_p$. For $p \in \{2, 3, 5, 7, 29\}$, the $\mathbb{F}_p$-algebra generated by the generators of S acting on L/pL has dimension 392 [C29] since the kernel of $E \to \mathrm{End}(L/pL)$ is pE, this says $A + pE = E$, and Nakayama's lemma gives $A_p = E_p$.

Now X is a lattice in W^2 stable under A, hence under E_p at every p. Since M is a maximal order, L_p is a free M_p-module of rank 7, so E_p is a full matrix ring over M_p^{op} and, by Morita equivalence, every E_p-stable lattice in W_p^2 is $L_p \otimes_{M_p} \Lambda_p$ for the left M_p-lattice $\Lambda_p = \mathrm{Hom}_{E_p}(L_p, X_p) \subset \mathcal{Q}_p^2$. Globally, $\Lambda = \mathrm{Hom}_S(L, X) = \mathrm{Hom}_E(L, X)$ (an A-linear map is E_p-linear at every p) is a left M-lattice in $\mathrm{Hom}_{\mathbb{Q}S}(W, X \otimes \mathbb{Q}) = \mathcal{Q}^2$ with localizations Λ_p, and $X = L \otimes_M \Lambda$ because both sides have the same localizations. For the dual, $X^{\#}$ is again S-stable, so $X^{\#} = L \otimes_M \Lambda'$ with $\Lambda' = \{\mu : \langle L\mu, L\lambda\rangle \subset \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$; as $\langle x\mu, y\lambda\rangle = \langle x\, h(\mu, \lambda), y\rangle$ and L is unimodular, $\mu \in \Lambda'$ iff $h(\Lambda, \mu) \subset M$, i.e. $\Lambda' = \Lambda^h$. Hence X is unimodular iff Λ is. An S-equivariant isometry $\sigma : X \to X'$ induces the M-linear bijection $\varphi \mapsto \sigma \circ \varphi$ from $\mathrm{Hom}_S(L, X)$ onto $\mathrm{Hom}_S(L, X')$, which preserves h because h is determined by the form of W^2; conversely an isometry $u : \Lambda \to \Lambda'$ gives the equivariant isometry $x\lambda \mapsto x\, u(\lambda)$ of $L \otimes_M \Lambda$ onto $L \otimes_M \Lambda'$. These constructions are mutually inverse, which gives the bijection of classes and $C_{\mathrm{Aut}(X)}(S) \cong U(\Lambda)$.

Proof of (b). *One genus.* At a finite place v of K, $\mathcal{Q}_v \cong M_2(K_v)$ and $M_v \cong M_2(\mathcal{O}_v)$ ($\mathcal{Q}$ is ramified only at the infinite places). Under Morita equivalence, Hermitian M_v-lattices in $\mathcal{Q}_v^2$ correspond to $\mathcal{O}_v$ -lattices in K_v^4 with an alternating form, unimodular lattices to self-dual ones [GHY01, §9]. Any two self-dual alternating lattices are isometric (both have a symplectic basis), so all unimodular lattices $\Lambda \subset \mathcal{Q}^2$ form one genus. This is the genus of maximal integral lattices in the sense of [GHY01]: for an alternating form, an integral lattice N with $N \neq N^{\#}$ can be enlarged by any $x \in N^{\#} \setminus N$ (because $\langle x, x\rangle = 0$), so maximal integral lattices are self-dual, and self-dual lattices are maximal because an integral $N' \supseteq N$ satisfies $N' \subseteq N'^{\#} \subseteq N^{\#} = N$.

Mass. By [GHY01, Prop. 9.3], for a totally definite quaternion algebra over a totally real field K of degree d that is unramified at all finite places (their $S_D = \emptyset$), the genus of maximal integral lattices of rank m has mass

$$\sum_{\Lambda} \frac{1}{|U(\Lambda)|} = 2^{-md} \prod_{r=1}^{m} |\zeta_K(1 - 2r)|,$$

the sum over the isometry classes in the genus. Here $m = d = 2$, and $\zeta_K(-1) = 1/2$, $\zeta_K(-3) = 157/20$ (computed exactly from generalized Bernoulli numbers and compared with Siegel's formulas and with PARI [C29]), so the mass is $157/640$. In rank 1 the same formula gives $2^{-2}\ |\zeta_K(-1)| = 1/8 = 1/24 + 1/12$, the two classes M and I of rank-1 unimodular lattices with their unit groups of orders 24 and 12, which confirms the normalization.

Enumeration. Starting from $\Lambda_0 = M \oplus M$, Kneser neighbours at the prime $\mathfrak{p} = \left(\left(7 + \sqrt{29}\right)/2\right)$ of norm 5 are generated by a random walk with a fixed seed; each neighbour is checked to be an M-module and unimodular. Isometry between two lattices is decided with PARI's `qfisom` and $|U(\Lambda)|$ with `qfauto`, applied to the nine integer-valued forms $\mathrm{Tr}_{K/\mathbb{Q}}\ \mathrm{trd}(m_k h)$, m_k running over 1 and a $\mathbb{Z}$-basis of M (a $\mathbb{Z}$-linear map preserving all nine is M-linear and preserves h). After 62 neighbours, 8 classes

with $\sum 1/|U(\Lambda_j)| = 157/640$ have been found [C29] by the mass formula there are no others. The classes are numbered by their invariants ($|U|$ and the numbers of vectors with Tr trd $h(\lambda,\lambda) \le 14$), which are distinct. The orders $|U(\Lambda_j)|$ are recounted independently as the numbers of S-equivariant automorphisms of X_j by exact enumeration, as in §8 [C31]. Each $X_j = L \otimes_M \Lambda_j$ is distributed by a reduced Gram matrix and by a basis in the coordinates of (3.1); the check verifies that the Gram matrix is even of determinant 1, that the basis realizes it, and that the generators of S act on X_j [C31].

Proof of (c). The lattices $M \oplus M$, $I \oplus I$ and $M \oplus I$ are unimodular. An element $1 \in \mathcal{O}_K$ is not a sum of two nonzero totally positive elements of $\mathcal{O}_K$, so the columns of an element of $U(M \oplus M)$ have exactly one nonzero entry, of reduced norm 1; hence $U(M \oplus M)$ is monomial of order $2 \cdot 24^2 = 1152$, and likewise $|U(I \oplus I)| = 2 \cdot 12^2 = 288$ and $|U(M \oplus I)| = 24 \cdot 12 = 288$ (no permutation, as $M \ncong I$). A vector λ with $h(\lambda,\lambda) = 1$ in a rank-1 unimodular lattice J gives $J = M\lambda \cong M$, so $M \oplus I$ has exactly 24 such vectors and $I \oplus I$ none. Comparing with Table S10 (column n_1 and $|U|$) identifies $\Lambda_0 = M \oplus M$, $\Lambda_2 = I \oplus I$, $\Lambda_6 = M \oplus I$, hence $X_0 = L \perp L$, $X_2 = \tilde{L} \perp \tilde{L}$, $X_6 = L \perp \tilde{L}$, of minimum $\min L = \min \tilde{L} = 6$. (In Table 2, the classes of f containing an element of C are these orthogonal sums: $L_u \cong L_1$ by Proposition 3.3(b), and $(x,v) \mapsto ((x+v)/2, (v-x)/2)$ maps $L_1 = \{(x,v) : v \equiv x \pmod 2\}/\sqrt{2}$ isometrically onto the orthogonal sum of two copies of the base lattice.) The isometries $X_7 \to L_{112}$ and $X_3 \to$ gluing A are explicit [C31], and gluing B $\cong$ gluing A [C28]. Vectors of norm 8 in X_3, X_4 and X_5 are distributed and checked in integer arithmetic [C30, C31] they were found in the fixed sublattices of elements su with $s \in S$ and $u \in U(\Lambda_j)$. Finally, X_1 is unimodular with $\min X_1 \ge 8$ by the rank-one sieve: every $\lambda \in \Lambda_1$ with $h(\lambda,\lambda) = c$ embeds (L, c) (the form scaled by c) into X_1, and the values c that occur for Tr trd $c \le 14$ are $2, 3, \pi, \bar{\pi}$ with $\pi = \left(7 + \sqrt{29}\right)/2$; the lattices (L, π) and $(L, \bar{\pi})$ have minimum 10 since they embed in $X_7 = L_{112}$. The decision between 8 and 10 for X_1 by the method of §§5–6, applied to the gluing of L and $\tilde{L}$, is in progress and will be reported in a later version.

Proof of (d). Let $\sigma : X_j \to L_{112}$ be an isometry and $S' = \sigma S \sigma^{-1} \subset \mathrm{Aut}(L_{112})$. By Theorem B(c), S (acting diagonally) is normal in $\mathrm{Aut}(L_{112})$ with quotient of order 48; the image of $S' \cong \mathrm{SL}_2(29)$ in this quotient is a quotient of $\mathrm{SL}_2(29)$ of order at most 48, hence trivial, so $S' \subseteq S$ and $S' = S$. Then σ conjugates $C_{\mathrm{Aut}(X_j)}(S) \cong U(\Lambda_j)$ onto $C_{\mathrm{Aut}(L_{112})}(S)$, which has order 48 by Theorem B(a). Table S10 shows $|U(\Lambda_j)| \ne 48$ for $j \ne 7$. □

Remarks. (1) Among the five non-split classes, four are gluings modulo 2 (frames b or c): the two of Table 2 with minimum ≥ 8 (classes 3 and 7) and two gluings of L with $\tilde{L}$ (classes 1 and 4), which Table 2 does not contain because it glues two copies of one base lattice. Class 5 has no frame modulo 2 at all; it has an automorphism of order 5 commuting with S ($|U(\Lambda_5)| = 20$). Every non-split class has a frame modulo the prime $\mathfrak{p}$ of norm 5 [C30]. (2) The proof of (b) relies on the computer-algebra systems for the isometry tests and the automorphism groups (PARI) and on the mass formula of [GHY01]; the orders $|U(\Lambda_j)|$, the evenness and unimodularity of the X_j, the action of S, the vectors of norm 8 and the two isometries are re-checked in integer arithmetic with numpy only [C31]. The deduction of (a) and (d) from these facts is not formalized in Lean.